Both Cycles in One Space

The Spheres of Dharma and their polar reversal sharing one space: the toggle between two exact settings, why they exclude each other, and the interlock, both cycles cut into one solid of convex pieces



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Data, scripts, exact checks and renders are included with this record and at https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-16-interlock and studies/2026-10-16-polar-dharma; shared checks in studies/_lib).
The Spheres of Dharma, turning them inside out, the idea of both cycles in one space as a toggle, the question whether the ends could be bent to give both settings, the interlock as one real solid, the fixed viewing bearings and the names of the pieces are DICTO's. Computation, exact arithmetic and renders were done with Claude (Anthropic) as a research assistant.

Abstract. The Spheres of Dharma nest six icosahedral solids, with the Hecatodihedron as an in-between stage; their polar reciprocal is a second exact cycle in which every nesting is reversed (Inverted Shells). We put both cycles in one space. The sphere of reciprocation is then the one free choice, and polarity with the cycle's φ^3 repeat gives mirror spheres every $\varphi^{3/2}$ in radius. At $\rho^2 = \varphi^2/4$ the two cycles share the icosahedron and dodecahedron; at $\rho^2 = 3\varphi^2/4$ they share the icosidodecahedron and triacontahedron; no sphere gives both, because the two settings differ by a factor 3, which is not a power of φ, and the loose ends of the cycle cannot be bent far enough to close the gap. Cutting every piece of one cycle by every piece of the other gives one solid of convex pieces, exact in $\mathbb{Q}(\sqrt{5})$ with remainder 0 in every piece and layer: per turn, 536 pieces of 12 kinds in the first setting and 502 pieces of 11 kinds in the second. The setting moves where the two cycles cut each other: the first cuts the outer roofs hardest and leaves the dodecahedron columns whole; the second leaves the roofs, pyramids and tetrahedra whole and cuts the columns. The innermost layer is cut in both.

1. Two cycles

The Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833) is an exact, endlessly repeating nesting of six icosahedral solids: icosahedron ⊂ icosidodecahedron ⊂ dodecahedron ⊂ truncated icosahedron ⊂ rhombic triacontahedron ⊂ icosahedron φ^3 larger, with the DICTO Hecatodihedron (DICTO DOI: 10.5281/zenodo.23360858) as the closed in-between stage of its fourth layer (stage 7 in *Capturing the In-betweens*, DICTO DOI: 10.5281/zenodo.23345502). Taking that stage as a seventh station and reciprocating every station gives the reversed cycle of *Inverted Shells* (DICTO DOI: 10.5281/zenodo.23405029): dodecahedron ⊂ icosidodecahedron ⊂ Ogdoecontahedron ⊂ pentakis dodecahedron ⊂ icosahedron ⊂ rhombic triacontahedron ⊂ dodecahedron φ^3 larger, the Ogdoecontahedron (DICTO DOI: 10.5281/zenodo.23404688) being the one new solid. DICTO asked whether the two cycles can share one space, and whether together they make one real solid.

Which closed stages count as stations is a choice. The truncated icosahedron → triacontahedron layer has two convex in-between stages: the Hecatodihedron (stage 7, on the main route) and the icosahedron 3·I (the truncated icosahedron with its 12 pentagonal caps restored, reached by skipping the Shields). Neither contains the other (float check), so at

most one can be a station. *Inverted Shells* chose the Hecatodihedron, and that choice produced the Ogdoecontahedron; the six published stations alone reciprocate to classical solids only. We keep the seven stations of *Inverted Shells*; the line-up with 3·I is left to a separate study.

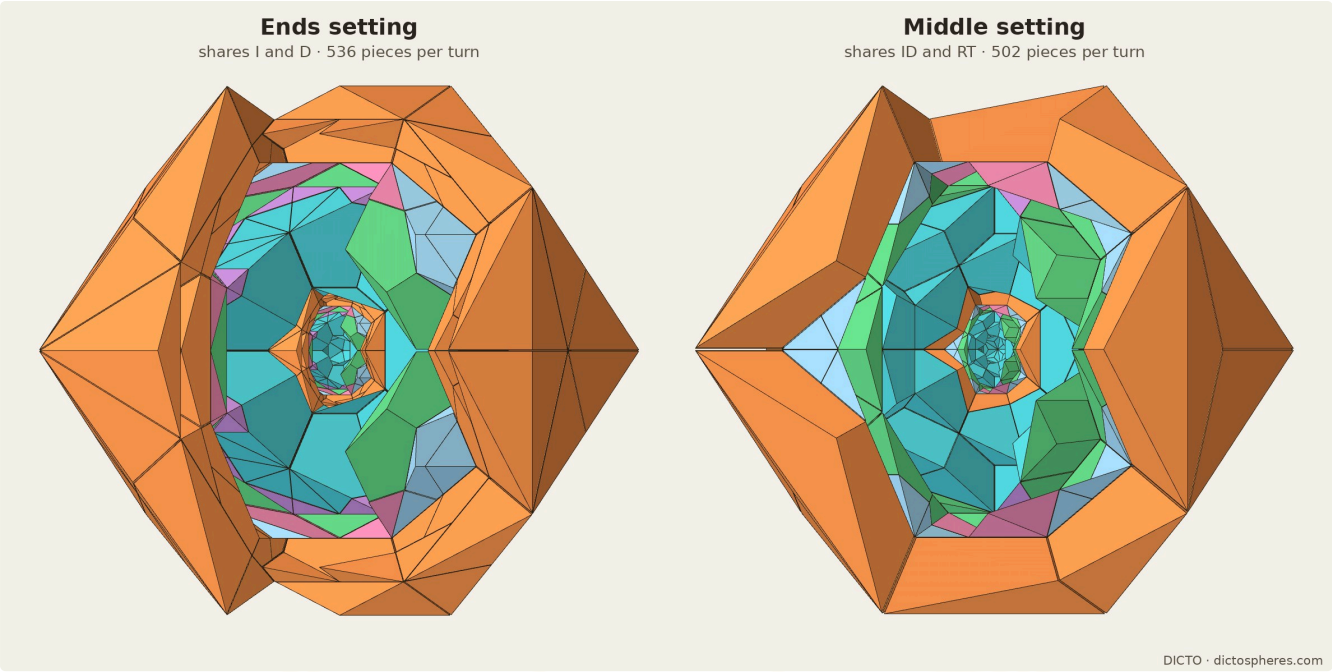


Figure 1. Both cycles as one solid: the interlock in its two settings, a slice removed so that you look inside; the next turn in sits at the centre (the same pieces scaled by φ^{-3}, as the nest repeats). Colours are the six layers of the original cycle; every seam is a cut by the reversed cycle. Ends setting: the outer roofs are cut finely. Middle setting: the roofs are whole and the cutting moves inward. Drawn from exact pieces.

2. The toggle

The sphere of reciprocation is the one free choice. Polarity in a sphere of radius ρ sends radius r to ρ^2/r, so in log-radius it is a reflection about log ρ; the cycle's φ^3 repeat is a translation. Together they give mirror spheres every $\varphi^{3/2}$ in radius: the union of the two cycles reads the same outward and inward, a palindrome in scale, and the polarity swaps the two cycles.

- **Ends setting**, $\rho^2 = \varphi^2/4 = (3 + \sqrt{5})/8$, mirror sphere radius $\varphi/2$: the toggle sphere sits at the geometric mean of the two radii it pairs, ρ^2 = (corner radius of I) × (face distance of D), and the polarity sends the original icosahedron to the original dodecahedron, so the two cycles share I and D. Five of the reversed turn's seven stations lie inside the original icosahedron, the fifth being that icosahedron; its last layer, triacontahedron → dodecahedron, reaches out to the original dodecahedron and holds about 93% of the turn's volume (float).
- **Middle setting**, $\rho^2 = 3\varphi^2/4$, mirror sphere radius $\sqrt{3}\cdot\varphi/2$: they share the icosidodecahedron and the triacontahedron instead. The reversed cycle is then exactly 3 times its ends-setting size, so its icosahedron is exactly 3·I: the original truncated icosahedron with its caps restored, an in-between stage of the original cycle though not one of its seven stations (checked in $\mathbb{Q}(\sqrt{5})$).
- **Never both.** A station can be shared only if its polar is also a station: of the seven, only I ↔ D and ID ↔ RT pair up this way (the truncated icosahedron and the Hecatodihedron go to the pentakis dodecahedron and the Ogdoecontahedron, which are not original

stations). Moving from one setting to the other scales ρ^2 by 3, while the cycle only repeats under powers of φ^3; 3 is not a power of φ (for any integer $k \neq 0$, $\varphi^k = (L_k + F_k\sqrt{5})/2$ with Lucas and Fibonacci numbers, and $F_k \neq 0$ makes it irrational), so no sphere gives both. Nor can the loose ends be bent to allow it: the original icosahedron can be scaled only within $[(45\sqrt{5} - 99)/2, 1 + 2\sqrt{5}/5] \approx [0.8115, 1.8944]$ before the nest breaks (the polar form of the exact bend range of *Inverted Shells*, §4), and sharing both pairs would need a scale of $3/\varphi^3 \approx 0.708$ or 3.

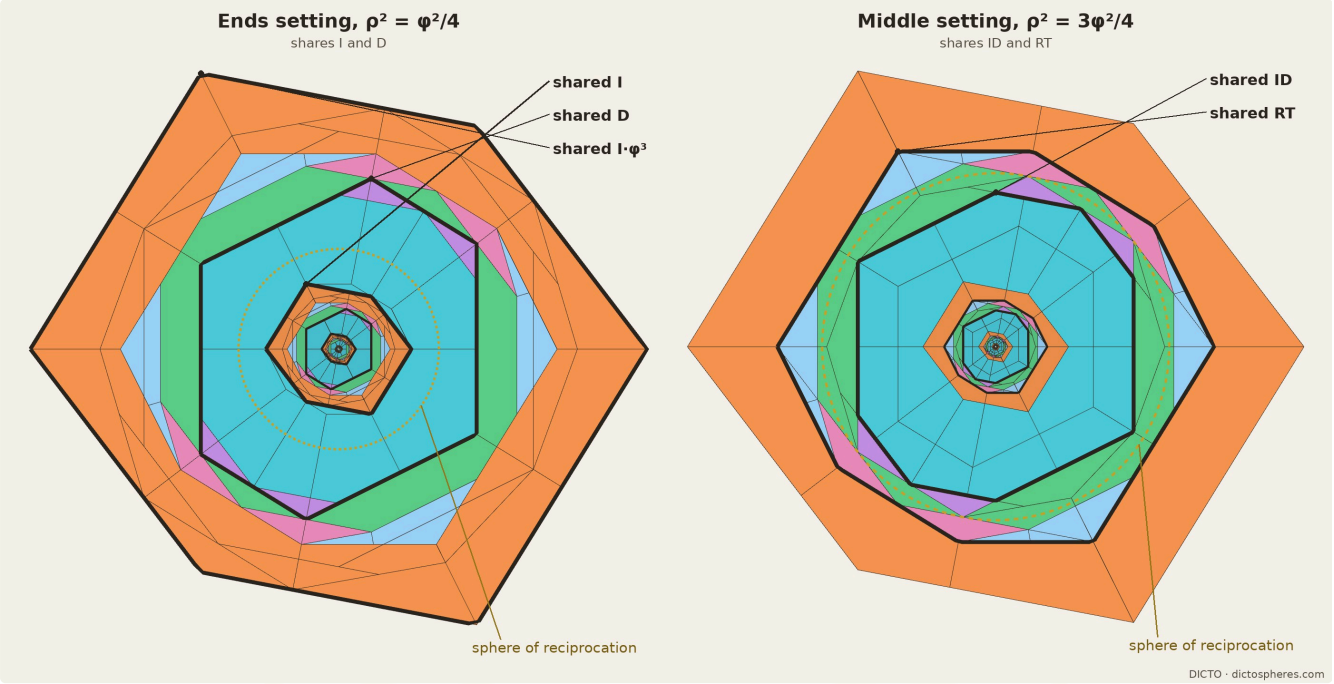


Figure 2. Both cycles as one solid, sliced clean through the centre and seen front on, like tree rings: the cut runs through a mirror plane of the icosahedral symmetry, every piece's cut face is coloured by its original layer, and the rings repeat inward turn after turn (the nest repeats under φ^3). Thick ink marks the stations the two cycles share; the dotted gold ring is the sphere of reciprocation, and where it sits decides which rings coincide. (a) $\rho^2 = \varphi^2/4$: I and D shared (and $I \cdot \varphi^3$, one turn out). (b) $\rho^2 = 3\varphi^2/4$: ID and RT shared. No sphere gives both. Drawn from exact pieces.

3. The interlock: both cycles as one solid

The reversed cycle was cut into convex pieces in *Inverted Shells*, by cones from the centre over the faces of each layer's inner solid (the Kion construction of *Theaetetus*, DICTO DOI: 10.5281/zenodo.23378070): 154 pieces in one turn. We cut the original cycle the same way, giving 174 pieces in one turn (these are new here, not the Dharma's own 237 pieces). We intersect every piece of one original turn with every piece of the reversed turns that cover it (turns 0 and 1 in the ends setting, −1 and 0 in the middle setting). Every intersection is convex. Every corner was then solved exactly, as the meeting point of three exact planes, after float points lying only on edges were dropped by a rank test on the plane normals. In both settings every original piece is exactly the sum of its parts, and every layer exactly the sum of its interlock pieces, with remainder 0 in $\mathbb{Q}(\sqrt{5})$. Kinds are counted twice, by exact volume and by symmetry orbit of the exact corners under the full icosahedral group (rotations and reflections, so a piece and its mirror image are one kind), and the two counts agree. All counts are per turn of the endless nest, from I to $I \cdot \varphi^3$. DICTO names the two sets as one family from the Inverted Shells project: **DICTO Interlock set 1 (Dharma-Ends)** and **DICTO Interlock set 2 (Dharma-Middle)**. Each piece is named by its set and its number in that set, largest first, followed by its shape: piece I1·1 (12f) is the largest piece of set 1.

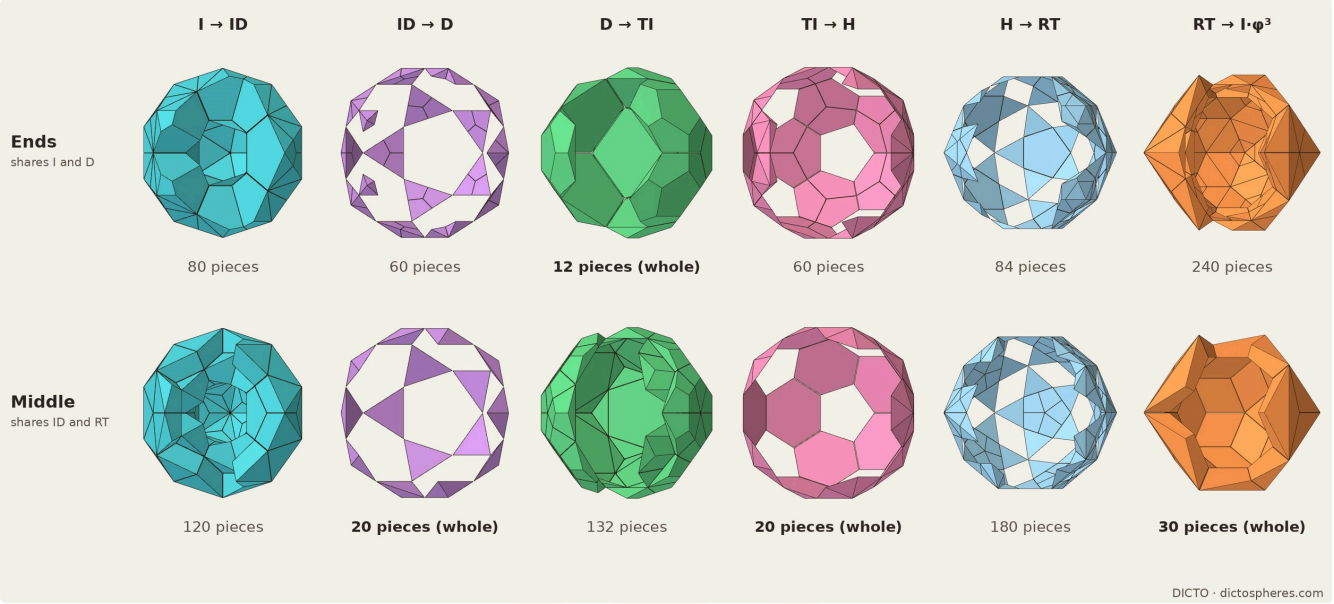


Figure 3. Each layer of one turn alone, a slice removed, in the ends setting (top) and the middle setting (bottom), with its exact number of interlock pieces (Table 3). Layers are scaled to fit their tiles. The two settings leave alternate layers whole. Drawn from exact pieces.

4. Interlock set 1 (Dharma-Ends): 536 pieces, 12 kinds

Several original layers lie wholly inside one reversed layer, so their pieces pass through unchanged; the cutting happens mainly in the outer layer, where the triacontahedron → I·φ³ roofs are cut by four reversed layers into 240 pieces.

Kind	Count	Original layer	Reversed layer	Corners	Faces (number × sides)	Volume (exact)
I1·1 12f	12	D → TI	D → ID	15	5 × 3, 7 × 5	-25/24 + 2/3√5
I1·2 TBP	60	RT → I·φ³	PD → I	5	6 × 3	145/1368 + 97/2052√5
I1·3 7f	60	I → ID	RT → D·φ³	9	3 × 3, 1 × 4, 3 × 5	83/576 − 41/2880√5
I1·4 7f	60	RT → I·φ³	D → ID	9	2 × 3, 3 × 4, 2 × 5	-11221/5184 + 15629/15552√5
I1·5 12f	12	H → RT	D → ID	15	5 × 3, 7 × 5	550235/24624 − 245243/24624√5
I1·6 7f	60	RT → I·φ³	ID → O	9	3 × 3, 1 × 4, 3 × 5	151655/32832 − 602341/295488√5
I1·7 QPy	60	RT → I·φ³	O → PD	5	4 × 3, 1 × 4	41/2592 + 55/7776√5
I1·8 PPy	60	TI → H	D → ID	6	5 × 3, 1 × 5	-11/8 + 5/8√5
I1·9 PPy	12	H → RT	ID → O	6	5 × 3, 1 × 5	-3715/162 + 1663/162√5
I1·10 Tet	60	H → RT	D → ID	4	4 × 3	-93/152 + 21/76√5
I1·11 QPy	60	ID → D	RT → D·φ³	5	4 × 3, 1 × 4	1/576 + 1/576√5
I1·12 Tet	20	I → ID	I → RT	4	4 × 3	-11/48 + 5/48√5

Table 1. Interlock set 1, the ends setting. Layers: I icosahedron, ID icosidodecahedron, D dodecahedron, TI truncated icosahedron, H Hecatodihedron, RT rhombic triacontahedron, O Ogdoecontahedron, PD pentakis dodecahedron; φ³ the next turn out. Corners and faces from the cleaned float corners; volumes exact. Pieces numbered largest first; shapes: 12f and 7f give the number of faces; TBP triangular bipyramid, PPy pentagonal pyramid, QPy four-sided pyramid, Tet tetrahedron.

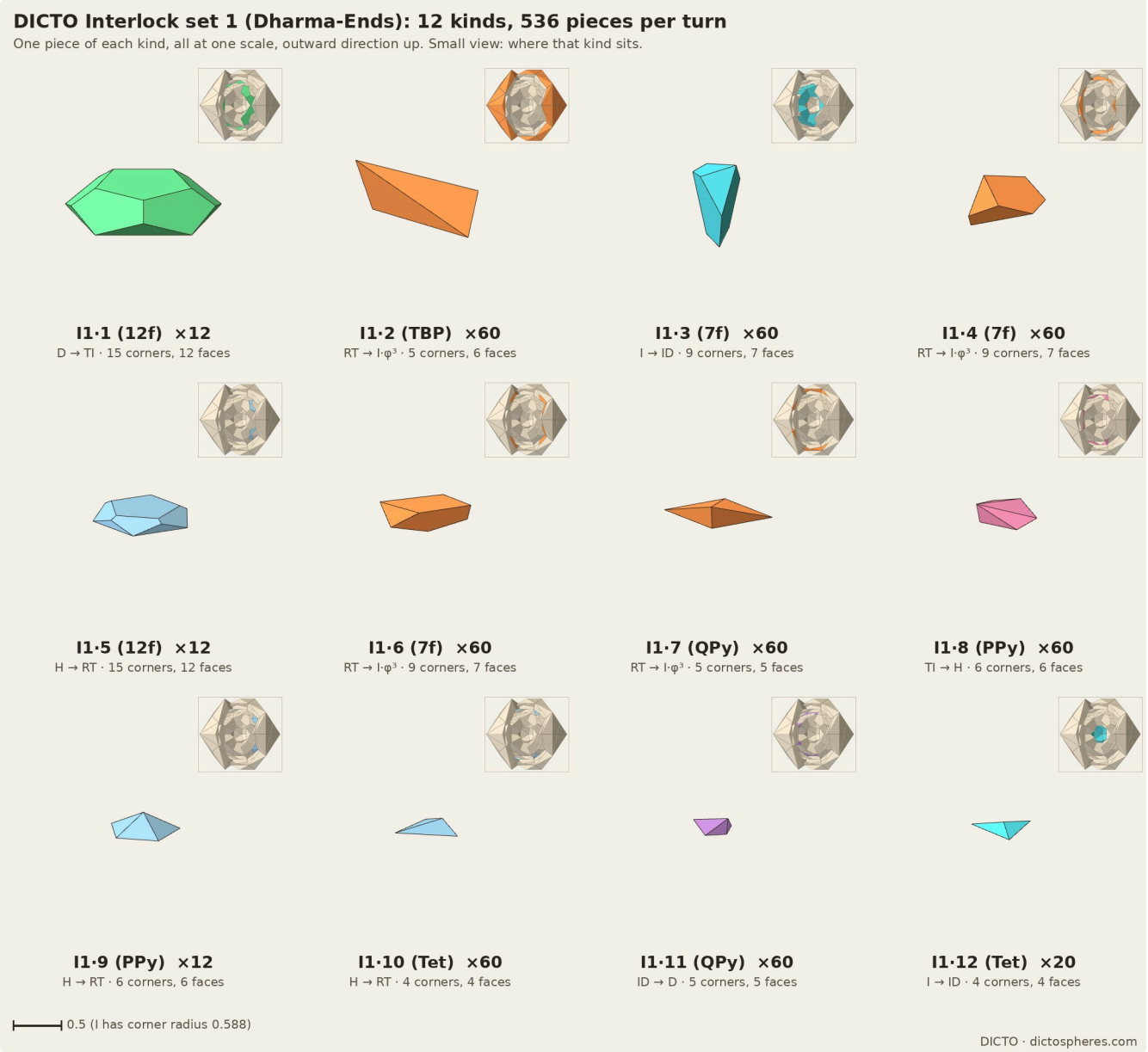


Figure 4. The parts of Interlock set 1 (Dharma-Ends): one piece of each of the 12 kinds, all at one scale with a scale bar, outward direction up, with its count; the small view shows where that kind sits (in colour, the rest pale). Drawn from exact pieces; exact volumes in Table 1.

5. Interlock set 2 (Dharma-Middle): 502 pieces, 11 kinds

At $\rho^2 = 3\varphi^2/4$ the reversed cycle is exactly 3 times its ends-setting size, so its pieces are the ends pieces scaled by 3. Here the outer roofs pass through whole: the 30 roofs (RT → I·φ³), the 20 truncated icosahedron → Hecatodihedron pyramids and the 20 icosidodecahedron → dodecahedron tetrahedra lie each inside one reversed piece. The cutting moves inward: the dodecahedron → truncated icosahedron columns are cut into 132 pieces, the Hecatodihedron → triacontahedron layer into 180 and the innermost layer, icosahedron → icosidodecahedron, into 120.

Kind	Count	Original layer	Reversed layer	Corners	Faces (number × sides)	Volume (exact)
I2·1 7f	30	RT → I·φ³	RT → D·φ³	8	2 × 3, 5 × 4	371/72 − 47/24√5
I2·2 11f	12	D → TI	ID → O	11	5 × 3, 5 × 4, 1 × 5	625/10944 + 289/10944√5
I2·3 7f	60	I → ID	D → ID	8	2 × 3, 5 × 4	-1585/576 + 3643/2880√5
I2·4 HPy	20	TI → H	I → RT	7	6 × 3, 1 × 6	-33/8 + 15/8√5
I2·5 7f	60	D → TI	PD → I	8	2 × 3, 5 × 4	-101/456 + 12/95√5
I2·6 6f	60	I → ID	RT → D·φ³	7	3 × 3, 2 × 4, 1 × 5	203/72 − 56/45√5
I2·7 Tet	20	ID → D	O → PD	4	4 × 3	1/192 + 1/192√5
I2·8 QPy	60	H → RT	PD → I	5	4 × 3, 1 × 4	-1/24 + 1/40√5
I2·9 Tet	60	H → RT	I → RT	4	4 × 3	-93/152 + 21/76√5
I2·10 Tet	60	D → TI	O → PD	4	4 × 3	1/576 + 1/576√5
I2·11 Tet	60	H → RT	I → RT	4	4 × 3	-23/304 + 11/304√5

Table 2. Interlock set 2, the middle setting, as Table 1. Further shapes: 11f and 6f give the number of faces; HPy hexagonal pyramid.

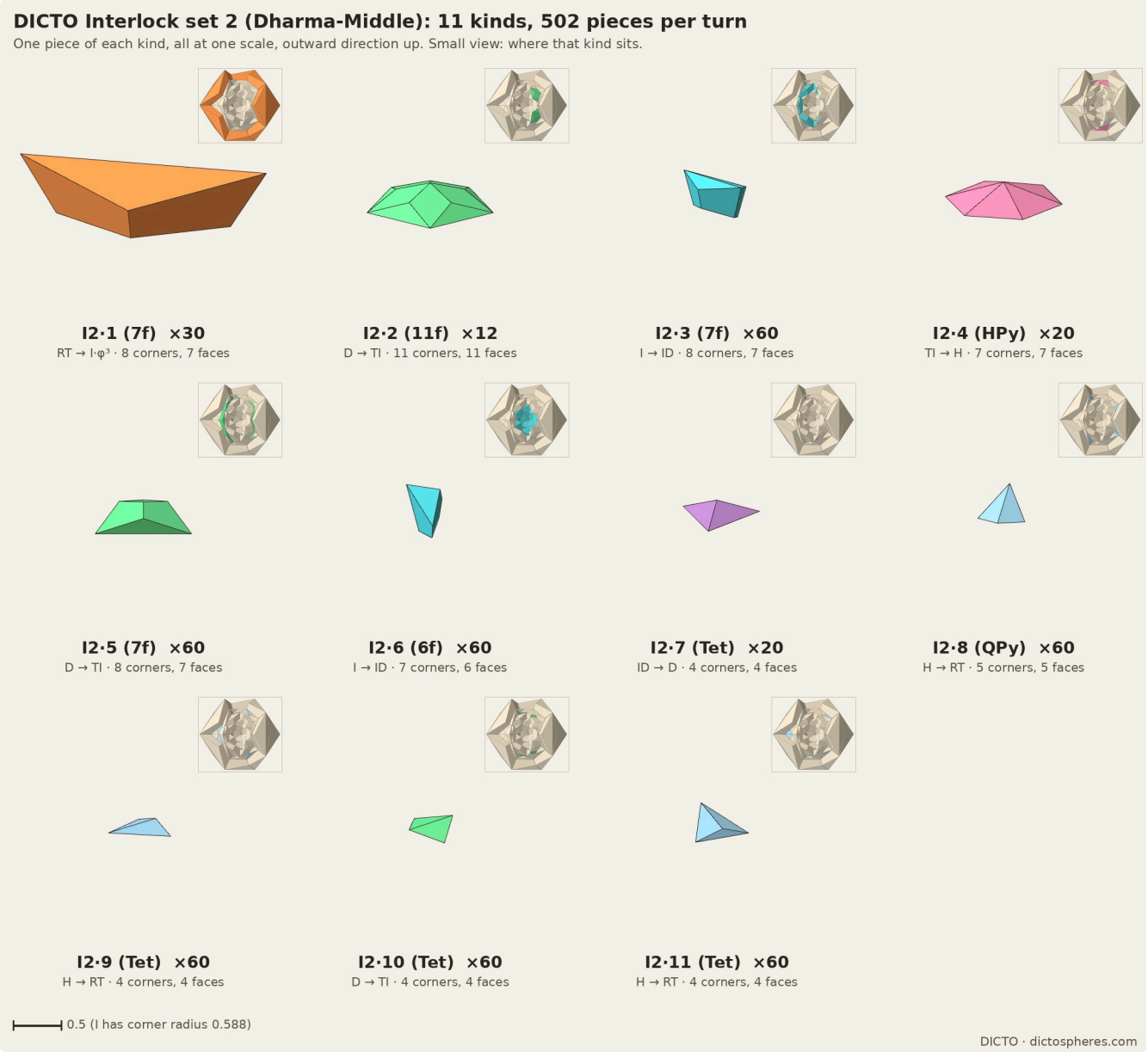


Figure 5. The parts of Interlock set 2 (Dharma-Middle), as Figure 4; exact volumes in Table 2. The 20 ID → D tetrahedra (I2·7) lie hidden behind other pieces in the small view.

6. What the toggle moves

- **Where the cycles cut each other.** The two settings leave alternate layers whole: the ends setting only the dodecahedron columns, the middle setting the tetrahedra, the pyramids and the roofs. The innermost layer is cut in both, harder in the middle setting (Table 3, Figure 3).
- **One piece in both.** The 60 small Hecatodihedron → triacontahedron tetrahedra, $(-93/152 + 21\sqrt{5}/76)$ each, pass through whole in both settings (I1·10 = I2·9).
- **Equal volume, different shape.** I1·11 and I2·10 both have volume $(1 + \sqrt{5})/576$ but different shapes (5 and 4 corners).

Original layer	Pieces before cutting	Ends setting	Middle setting
I → ID	20	80	120
ID → D	20	60	20 (whole)
D → TI	12	12 (whole)	132
TI → H	20	60	20 (whole)
H → RT	72	84	180
RT → I·φ³	30	240	30 (whole)
One turn	174	536	502

Table 3. Interlock pieces per original layer, per turn (exact counts, from Tables 1 and 2).

7. Relation to known work, and status

Known and credited: nested solids separated by spheres (Kepler 1596; here a sphere instead *reciprocates* one nest into another); polar reciprocation and the reversal of inclusion (convex geometry; Firsching 2015). Reciprocating in the midsphere gives the classical compound in which the icosahedron and dodecahedron cross; here the sphere is chosen so that they nest and coincide with stations. Further: the common refinement of two convex dissections by pairwise intersection, a standard construction; nested icosahedral shells in clusters, viruses and fullerenes (Takakura et al. 2007; Dechant, Wardman, Keef and Twarock 2014; Twarock, Valiunas and Zappa 2015); the cone cut (Kion construction, *Theaetetus*). **Not found** in our reference library and web searches (16 October 2026; printed catalogues not read): a cycle of nested solids and its polar reversal placed in one space; the two exact toggle settings and the proof that they exclude each other; the exact interlock dissections and their kinds.

8. Reproducibility

In studies/2026-10-16-interlock: `chains.py` (both chains in one frame), `interlock.py` (the pairwise intersections, float; arguments for the middle setting), `interlock_exact.py` (exact corners, sums and kinds → `interlock-ends-exact.txt`, `interlock-middle-exact.txt`), `interlock_kinds.py` (Tables 1 and 2 → `interlock-kinds.txt`), `interlock_kinds_render.py` (cleaned corners of one piece per kind) and `p4_render_data.py`, `render_p4.html`, `shoot_p4.mjs` (Figures 1, 3, 4 and 5), `p4_xray_data.py` (stations) and `shoot_p4_slice.mjs` (Figure 2) and `p4_compose.py`; `p4_figures.py` draws the record's flat diagrams, `middle_icosa_3I.py` (the reversed icosahedron of the middle setting is exactly 3·I → `middle-icosa-3I.txt`). Checks: `python3 studies/_lib/test_geomcheck.py`.

References

1. J. Kepler, *Mysterium cosmographicum*, Tübingen, 1596.
2. M. Firsching, Computing maximal copies of polyhedra contained in a polyhedron, *Experimental Mathematics* 24 (2015) 98–105, doi:10.1080/10586458.2014.956374.
3. H. Takakura, C. P. Gómez, A. Yamamoto, M. de Boissieu, A. P. Tsai, Atomic structure of the binary icosahedral Yb–Cd quasicrystal, *Nature Materials* 6 (2007) 58–63, doi:10.1038/nmat1799.
4. P.-P. Dechant, J. Wardman, T. Keef, R. Twarock, Viruses and fullerenes: symmetry as a common thread?, *Acta Cryst. A* 70 (2014) 162–167, doi:10.1107/S2053273313034220.
5. R. Twarock, M. Valiunas, E. Zappa, Orbits of crystallographic embedding of non-crystallographic groups and applications to virology, *Acta Cryst. A* 71 (2015) 569–582, doi:10.1107/S2053273315015326.
6. DICTO, Capturing the In-betweens: a Study of Transitional Polyhedral Shells, DICTO DOI: 10.5281/zenodo.23345502.

7. DICTO, Spheres of Dharma: a second endless golden cycle of the icosahedral solids, through the icosidodecahedron, the fullerene cage and the rhombic triacontahedron, DICTO DOI: 10.5281/zenodo.23310833.
8. DICTO, Theaetetus: an icosahedron held exactly by its Cleo-Icos, grown to the DICTO Hecatodihedron in two exact stories, DICTO DOI: 10.5281/zenodo.23378070.
9. DICTO, The DICTO Truncated-Icosahedral Hecatodihedron, DICTO DOI: 10.5281/zenodo.23360858.
10. DICTO, The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688.
11. DICTO, Inverted Shells: The Spheres of Dharma Turned Inside Out, DICTO DOI: 10.5281/zenodo.23405029.